MILITARY SPECIFICATION

SEPARATOR, SERUM AND PLASMA, 100s

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a disposable device for the separation of serum and plasma from whole blood.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-T-36974	Tube, Blood Collecting, Vacuum, Non-Sterile.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6640

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2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Food and Drug Administration

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Food and Drug Administration, U.S. Department of Health, Education and Welfare, Washington, DC 20204.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein. Where material is not specified, it shall be suitable for the intended purpose.

3.1.2 Sealant. Sealant shall be a semi-solid mixture of silicone fluid and silica (see 6.4).

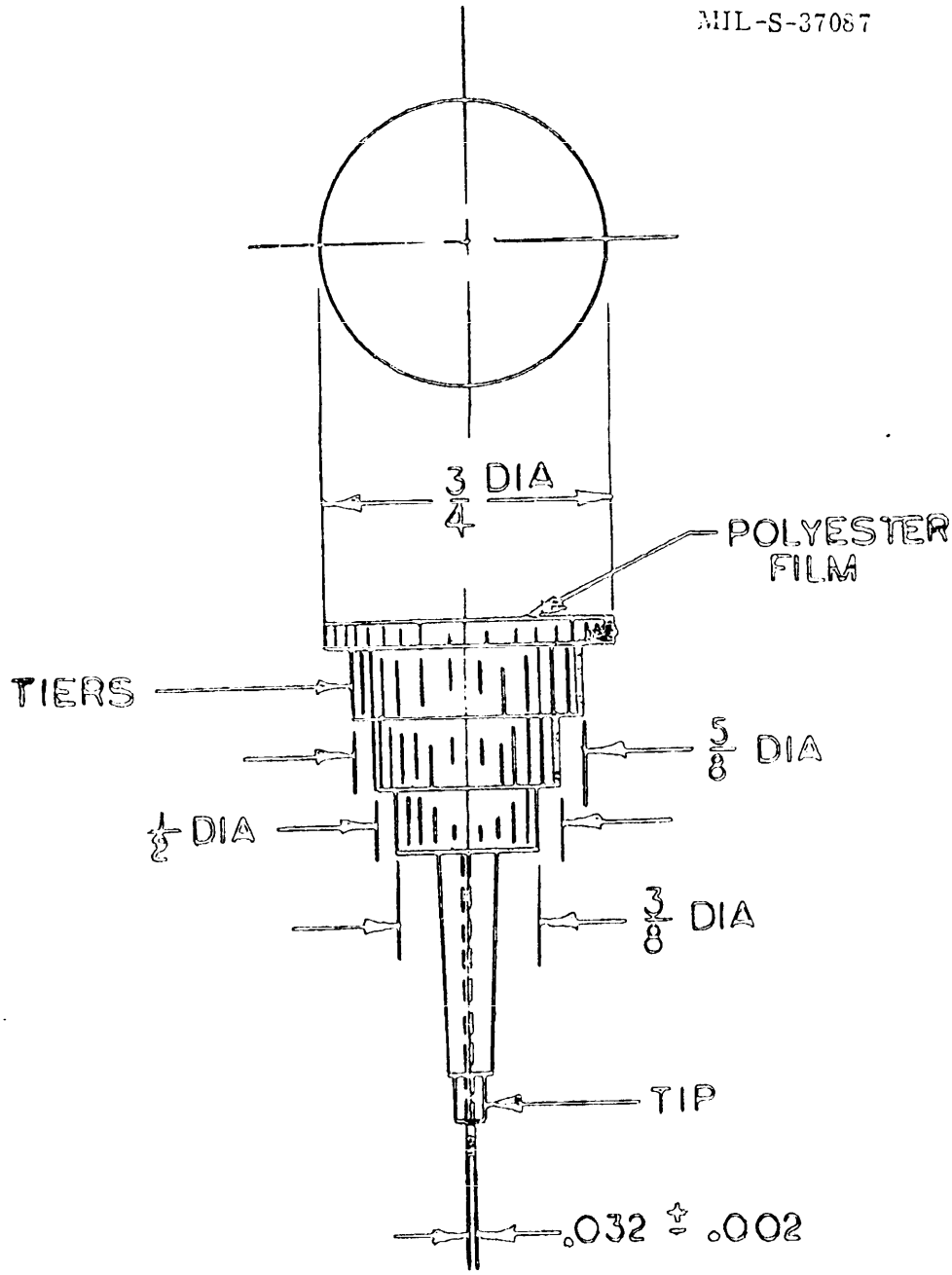
3.2 Construction. Separator shall consist of a tiered polyethylene funnel and sealant. Sealant shall be in the tiered portion of the funnel. Separator shall act as a cover for blood collecting tubes during centrifugation and shall be in accordance with Figure 1. Variations from tier and top rim diameters shown in figure shall not be cause for rejection, provided these diameters are such that the tiered portion or the top rim covers the 13- and 16-mm. diameter tubes listed in 6.3 when tested as specified in 4.4.1 and 4.4.2.

3.3. Performance. The separator shall act as a cover, the polyester film shall remain intact and the sealant shall vacate the two top tiers when tested as specified in 4.4.1 and 4.4.2. Evacuation of more than the two top tiers shall not be cause for rejection. Sealant shall be on top of the solution after being tested in accordance with 4.4.1 and shall sink to the bottom of the tube when tested as specified in 4.4.2.

3.4 Weight. Average of the sealant in the separator shall be 1.5 $\pm$ .2 grams when tested as specified in 4.4.3.

3.5 Instructions. Instructions for use shall be furnished with each unit package.

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DIMENSIONS ARE IN INCHES

FIGURE 1

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3.6 Workmanship. Workmanship shall be first class throughout. Separator shall be free of defects listed in Table III and any other defects which may detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Table I and MIL-STD-105.

Table I For examination

	Unit of product	Inspection level	AQL (percent defective)
For visual examination			
Major A	Package	II	1.0
Major B	* Separator	II	1.0
Minor	Package	II	2.5
For dimensional examination	Separator	S-1	2.5

* No more than 20 separators shall be taken from any unit package.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Table II and MIL-STD-105. Unit of product shall be one separator. Acceptance number shall be zero.

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Table II For tests

Characteristic	Requirement	Test procedure	Inspection level
Density	3.3	4.4.1	S-1
		4.4.2	S-1
Weight	3.4	4.4.3	*

*Ten samples shall be tested from each lot.

4.3 Examination. The separator shall be examined to determine compliance with all requirements contained in this specification. Examination shall be in accordance with, but shall not be limited to, the defects listed in Table III.

Table III Classification of defects

Category	Defect
Major A	
101	Instructions missing.
102	Instructions not legible.
103	Package not containing specified number of separators.
Major B	
151	Polyester film not secured.
152	Polyester film not free of tears.
153	Outside surface of tip not free of sealant.
154	Tiered portion does not contain sealant. (Portion not containing sealant is translucent; portion containing sealant is opaque.)
155	Funnel not free of deformation.
Minor	
201	Identification marking not legible.

4.3.1 Dimensional examination. The separator shall be examined for defects in dimensions. Any tip orifice not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Testing in 4.4.1 and 4.4.2 shall be such that an equal number of 13- and 16-mm. diameter blood collecting tubes is used (see 6.3).

4.4.1 Heavy solution. Sufficient salt, such as copper sulfate, shall be dissolved in water to form a solution which has a specific gravity of 1.055 at 25°Centigrade (°C.). Solution shall be put into a 13- or 16-mm. blood collecting tube to a point where the solution will cover the tip of the separator. Separator shall be observed

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to determine that there is sealant inside of all tiers of the funnel (see Table III). Separator shall be inserted in tube and the tube inserted in a centrifuge. Centrifuge shall be run for 10 minutes at a speed which produces a relative centrifugal force of 500 gravities.

4.4.2 Light solution. Sufficient salt, such as copper sulfate, shall be dissolved in water to form a solution which has a specific gravity of 1.035 at 25°C. Procedure described in 4.4.1 shall be repeated.

4.4.3 Weight. Tare weight of separator without sealant shall be determined. Separator with sealant shall be weighed; weight of sealant shall be calculated and shall be reported as the average weight of sealant in ten separators.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package, containing one hundred separators, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	24 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. One hundred separators shall be packaged in a molded or die-cut polyurethane, polystyrene or plastic tray in such a manner that each separator is contained in an individual cell. Each tray shall be inserted in a paperboard sleeve or box. The sleeve or box may be overwrapped in a clear plastic wrap.

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5.1.3.2 Intermediate package. Intermediate package shall be a snug-fitting box constructed in accordance with PPP-B-636, type CF, class domestic. The box shall be secured to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class, 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Special requirements as specified in the applicable table of PPP-B-636 shall apply for fiberboard boxes.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. In addition, fiberboard boxes shall conform to the special requirements specified in the applicable table of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129 and the requirements of the Federal Food, Drug and Cosmetic Act. The date of manufacture shall be shown in lieu of the date packed. Lot number and instructions for use shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59°-86°F.)."

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown in lieu of the date packed. Lot number and the legend, "STORE AT CONTROLLED ROOM TEMPERATURE (59°-86°F.)," shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. The date of manufacture shall be shown in lieu of the date packed. Lot number shall be shown. Marking shall include the legends, "STORE AT CONTROLLED ROOM TEMPERATURE (59°-86°F.)" and "MATERIAL MAY BE SHIPPED BY AIR IF PLACED IN PRESSURIZED COMPARTMENT", or similar statement.

6. NOTES

6.1 Intended use. The item covered by this specification is intended for use to prevent evaporation, to aid in separation of plasma and serum and to clean plasma and serum during centrifugation of blood.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Separator is used with the following blood collecting tubes which are in accordance with MIL-T-36974:

National Stock No.	Item Identification (tube diameter)
6630-00-145-1132	Tube, Blood Collecting, Vacuum, 5 ml. with Ammonium and Potassium Oxalate, 100's, 13-mm. tube

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National Stock No.	Item Identification (tube diameter)
6630-00-145-1137	Tube, Blood Collecting, Vacuum, 7 ml., without Anticoagulant, 100's, 13-mm. tube
6630-00-145-1143	Tube, Blood Collecting, Vacuum, 15 ml., without Anticoagulant, 100's, 16-mm. tube
6630-00-145-1527	Tube, Blood Collecting, Vacuum, 15 ml. with Lithium Oxalate, 100's, 16-mm. tube
6630-00-145-1533	Tube, Blood Collecting, Vacuum, 7 ml., with Disodium Edetate, 100's, 13-mm. tube
6630-00-145-1534	Tube, Blood Collecting, Vacuum, 7 ml., with Sodium Fluoride, 100's, 13-mm. tube
6630-00-250-4264	Tube, Blood Collecting, Vacuum, 5 ml., with Sodium Citrate, 100's, 13-mm. tube

6.4 Sealant fabricated of silica fluid and silica has been found to be inert when used with tubes listed in 6.3. Other sealants will be considered, provided detailed evidence is supplied which shows that the alternates do not interfere with separation (see 6.1) and with the determinations made on the separated plasma and serum.

6.5 The item covered by this specification is listed in the Federal Supply Catalog as follows:

National Stock No.	Item Identification
6640-00-397-2375	SEPARATOR, SERUM AND PLASMA, 100s

6.6 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet Military requirements.

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Army - MD
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Air Force - 03

Preparing activity:
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Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-S-37087

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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